

LASUSTECH PHY 208 – ACOUSTICS

LECTURE NOTE

NOISE POLLUTION

Unwanted sound is called noise. Pollution produced in the air due to noise is called noise pollution.

Noise sources: The principal sources of noise may be classified as surface transportation noise, aircraft noise, industrial noise, noise due to construction sites and noise at home.

There are three types of noises.

(a) Air-borne noise (b) Structure borne noise and (c) Inside noise. The prevention of the transmission of noise inside or outside the hall is called sound insulation.

(a) Air-borne noise: the noise carried by air that enters into the hall from outside through open windows, doors and ventilators is called air-borne noise.

Sound insulation is achieved for air-born by the following methods:

1. Very heavy glasses are used in doors, windows and ventilators
2. The windows and doors are arranged such that outside noises are prevented from entering the room.
3. Double doors with absorbing materials in between are used.
4. Openings for pipes and ventilators are avoided.

(b) Structure borne noise: structure borne noises are produced inside the hall itself and conveyed through the structure. Any loose fitting inside building vibrates and produce noises. Other activities such as drilling, hammering, operating machinery inside the building, moving of furniture etc., also produce structure-borne noise.

Sound insulation for the reduction of structure borne noise is done in the following ways.

1. By breaking the continuity of pipe lines etc. by interposing sound insulation materials.
2. By arresting vibrations of loose fittings
3. It can be done by insulating the machinery and having carpets, rubber sheets etc., on the floor.

(c) Inside noise: The noises which are produced inside the hall are called inside noises. They are produced due to machinery like Television sets, automatic washers, air conditioners, type writers etc, in the hall.

Following methods are used for sound insulation of inside noise.

1. The walls, floor and ceiling of the room are padded with sound absorbing materials.
2. Working machineries are placed on absorbent pads

3. Any engine inside the hall should be fitted on the floor with a layer of wood or felt between them.

Effects of Noise Pollution

1. Noise affects work efficiency.
2. It produces mental fatigue and irritation
3. A person's hearing becomes damaged when he is prolongedly exposed for a long time to sounds of high intensity (140 – 150 dB).
4. Noise produces a change in bodily activity. The results are indigestion, sleeplessness, nervousness and nausea.

ACOUSTICS OF BUILDINGS

The branch of Physics which deals with the design and construction of buildings with good acoustic properties is called “Acoustics of buildings.” It deals basically with (i) reverberation control, (ii) noise insulation and reduction and (iii) sound distribution and absorption.

The following are the requirements of a good auditorium.

1. The sound should be sufficiently loud and intelligible in every part of the hall.
2. Sound of each syllable should soon decay so that the succeeding syllable may be heard distinctly. There must be no confusion due to overlapping of syllables. This means that the auditorium must be free from excessive reverberation.
3. No echoes should be present.
4. There should not be undesirable focusing of sound in any part of hall. There should not be any zones of silence or regions of poor audibility anywhere in the hall.
5. Interference, reflection and resonance effects should be avoided.
6. All extraneous noises must be shut out as far as possible. The boundaries should be sufficiently sound proof to exclude extraneous noise.
7. There should be no echelon effect.
8. The quality of sound must be unaltered i.e., the relative intensity of the components of the complex sound must be preserved.

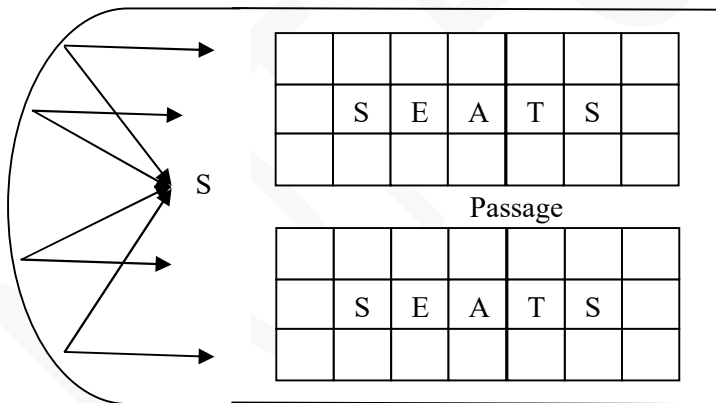


Fig. 1: The usual design of an auditorium or hall

It has a parabolic shape at the speaker's end S. As the speaker speaks, sound is reflected from the back almost as a parallel beam. So there is uniform distribution of sound intensity throughout the hall.

5.2 FACTORS AFFECTING THE ACOUSTICS OF BUILDINGS

The factors affecting the acoustics of a hall are:

- i. Reverberation time
 - ii. Presence of echoes
 - iii. Adequate loudness
 - iv. Focusing and interference effects
 - v. Resonances within the building
 - vi. Echelon effect
 - vii. Extraneous noises
 - viii. Inside noise.
- (i) **Reverberation.** Reverberation is one of the important single factors that affects the acoustics of a room or a hall.

The reverberation of sound in an auditorium is due to multiple reflections taking place at serious surfaces present within the auditorium.

If the reverberation time is too small, the sound dies away almost instantaneously and gives the hall a 'dead' effect. If the reverberation time is too long, each syllable continues to be heard even after the next syllable has been uttered. This makes the sound unintelligible. Therefore, the value of reverberation time is maintained at an optimum value. Rooms of different dimensions were used by Sabine to find the optimum reverberation time. For speech, the reverberation time should be about 1 to 2 seconds. For music, it is between 2 to 2.5 seconds.

To adjust the reverberation time to the best optimum value, the absorption of sound in the hall is increased in the following ways:

- a. Providing windows and openings
- b. Covering ceiling and walls with sound absorbing materials like felt, fibre board, celotex, asbestos etc.
- c. Using heavy curtains with folds
- d. Covering the floor with carpets
- e. By upholstering the seats so that the absorption is approximately the same with or without the audience.
- f. By having a good audience because one listener is equivalent to about 0.5m^2 area of an open window and
- g. By decorating the walls with pictures, maps etc.

- (ii) **Echoes:** Echoes can be avoided by covering the long distance walls and high ceiling with absorbent materials. This prevents the reflection of sound.
- (iii) **Adequate loudness:** Sufficient loudness in every portion of the hall is an important factor for satisfactory hearing. If the loudness of the sound is inadequate, the loudness can be increased by
 - (a) Using large sounding boards behind the speaker and facing the audience.
 - (b) By providing low ceiling for the reflection of sound towards the audience and
 - (c) By providing additional energy with the help of equipments like loud speakers.
- (iv) **Focusing:** if there is any concave surface in the hall, sound is concentrated at its focus region. There may be dead space at some other regions. Hence such surfaces must be avoided. If curve surfaces are present, they should be covered with absorbent materials.

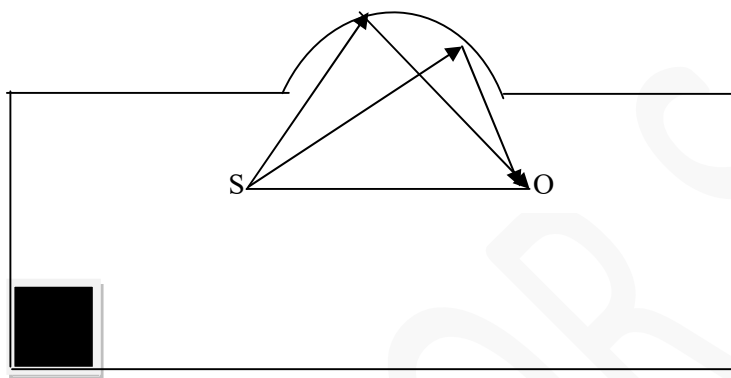


Figure 2:

In Figure 2, the observer at O receives sound from the speaker along the direct path SO . The observer also receives the sound waves after reflection from the ceiling. Thus the intensity of sound at O is comparatively higher than at other positions in the auditorium. There should be no interference of direct and reflected waves. Good absorbing materials avoid this defect.

Suppose the direct and the reflected waves are in opposite phase at O . This results in minimum intensity of sound at O . Further, the direct and the reflected waves may form a stationary wave pattern. This causes uneven distribution of sound intensity.

- (v) **Resonance Effects:** Window panels, sections of wooden portions which are loosely fitted start vibrating, creating extra sounds. If there is resonance for any audio frequency note, the intensity of the note will be entirely different from the intensity desired. Such resonant vibrations should be damped.
- (vi) **Echelon effect:** Suppose there is regular structure similar to a flight of stairs in the hall. The sound produced in front of such a structure may produce a musical note due to regular successive echoes of sound reaching the observer. Such an effect is called echelon effect. If the frequency of this note is

within the audible range, the listener will hear only this note prominently. The remedy to cover such surfaces with absorbing materials like carpets.

- (vii) **Extraneous noises:** They are classified as air borne and structure borne noises.
 - (a) **Air borne noise:** the noise which reaches the hall from outside through open windows doors and ventilators is called Air borne noise. It is reduced by the following methods.
 1. Using double doors and windows with separate frames and having insulating materials between them.
 2. Making the hall as an air-conditioned hall where the hall is completely closed.
 - (b) **Structure borne noise:** The noise which is conveyed through the structure of the building is called structure borne noise. This noise can be minimized by using double walls with air space between them.
- (viii) **Inside noise:** the noise which is produced inside the hall is called inside noise. It is produced by machinery like air conditioners, type writers etc., in the hall. This noise is minimized by the following methods.
 1. Covering the floor with carpet
 2. Walls, floor and ceiling should be provided with sound absorbing materials.
 3. Placing the working machineries like air-cooler or air conditioning machine on sound absorbent pads.
 4. The rotating or impacting machines must be fixed using a fixed bed.

SOUND DISTRIBUTION IN AN AUDITORIUM

Sound distribution describes how the sound pressure level varies with position in an enclosure. The design of an auditorium requires smooth decay and growth of sound. In an auditorium, the sound must be distributed or diffused over the whole area. To ensure these factors, acoustic treatment is given viz, scattering effect of objects, irregularities on the wall surfaces, fixing absorptive material on the walls etc. Model analysis with light rays, ultrasonic waves or ordinary audio frequency sound is used to study sound distribution. Graphical construction of first reflection of the sound waves at various cross-sections can also be used as in Figure 1. It is clear from the figure that the reflected sound is distributed evenly in the whole auditorium viz, the main floor and the balcony. This design enables an even distribution of sound intensity.

REVERBERATION

When a source produces sound waves inside a closed building, the waves are generally reflected by walls, ceiling and other materials present in the room. The intensity of the sound wave decreases at every reflection and finally the sound becomes inaudible. So the listener receives (i) direct waves and (ii) reflected waves due to multiple reflections, at the various surfaces. There is also a time gap between the direct wave

receives by the listener and the waves received by successive reflection. Due to this, the sound persists for sometime even after the source has stopped. This persistence of sound is called reverberation.

Definition of Reverberation

Reverberation is the persistence of sound in an enclosure due to multiple reflection of sound at the walls after the source has ceased to emit sound.

Definition of Reverberation time

The interval of time taken by sustained or continuous sound to fall in intensity to one millionth of its original value is called reverberation time (T) OR Reverberation time is defined as the time taken for the energy density to fall to one-millionth of its value before the cut off.

It is the time taken for the sound to fall below the minimum audibility after the source stopped sounding.

The reverberation time will depend on the size of the room or the auditorium, the nature of the reflecting material on the wall and the ceiling and the area of the reflecting surfaces.

Sabine's method of determining reverberation time

An organ pipe of frequency 512 Hz is sounded. At the same time, a stop watch is started. The time for the sound to become inaudible is measured. This gives the reverberation time.

Sabine's formula

Sabine's formula for reverberation time is

$$T = \frac{0.158v}{\sum A\alpha} \quad (1)$$

According to equation (1) The reverberation time is

- (i) Directly proportional to the volume of the auditorium;
- (ii) Inversely proportional to the area of the ceilings, area of the walls etc and
- (iii) Inversely proportional to the total absorption plus transmission through open surfaces.

Importance of Sabine's Formula

If the reverberation time is too small, the sound dies away almost instantaneously and gives the hall a 'dead' effect. If the reverberation time is too long, each syllable continues to be heard even after the next syllable has been uttered. This makes the sound unintelligible. Therefore, the value of reverberation time is maintained at an optimum value. Rooms of different dimensions were used by Sabine to find the optimum

reverberation time. For speech, the reverberation time should be about 1 to 2 seconds. For music, it is between 2 to 2.5 seconds.

ABSORPTION COEFFICIENT

First Definition: The sound absorption coefficient (α) of a material as the ratio of sound energy absorbed by it to the total energy incident on it.

$$\text{Absorption coefficient } (\alpha) = \frac{\text{Sound energy absorbed by the surface}}{\text{Total sound energy incident on the surface}}$$

Sabine chose 1 square metre of an open window as a standard unit of absorption since all sound waves falling on it pass through and can be said to be completely absorbed. This unit is called Sabine. It is also called open window unit (O.W.U).

Second Definition. The absorption coefficient of a material is defined as the ratio of sound energy absorbed by it to that absorbed by an equal area of an open window.

WORKED EXAMPLES

1. A hall of volume 5500 metre³ is found to have a reverberation time of 2.3 s. The sound absorbing surface of the hall has an area of 750 m². Calculate the average absorption coefficient.

Solution. Here, $V = 5500 \text{ m}^3$. $T = 2.3 \text{ s}$, $A = 750 \text{ m}^2$

Reverberation time, T (*Sabine's formular for reverberation time*)

$$\alpha = \frac{0.158V}{AT} = \frac{0.158 \times 5500}{750 \times 2.3} = \mathbf{0.504}$$

2. The volume of a room is 600 m³. The wall area of the room is 220 m², the floor area is 120 m² and the ceiling area is 120 m². The average sound absorption coefficient, (i) for the walls is 0.03; (ii) for the floor is 0.06 and (iii) for the ceiling is 0.80. Calculate the average sound absorption coefficient α and the reverberation time.

Solution. Average absorption coefficient.

$$\alpha = \frac{\sum \alpha A}{\sum A} = \frac{\alpha_1 A_1 + \alpha_2 A_2 + \alpha_3 A_3}{A_1 + A_2 + A_3} = \frac{0.03 \times 220 + 0.06 \times 120 + 0.8 \times 120}{220 + 120 + 120} = 0.2387$$

Total sound absorption of the room.

$$\alpha \sum A = 0.2387 \times 460 = 109.8 \text{ metric sabine}$$

Reverberation time,
$$T = \frac{0.158V}{\alpha \Sigma A} = \frac{0.158 \times 600}{109.8} = \mathbf{0.8634 \text{ s}}$$

3. A hall of volume 5500 metre³ is found to have a reverberation time of 2.3 s. The sound absorbing surface of the hall has an area of 750 m². Calculate the average absorption coefficient.

Solution. Here, $V = 5500 \text{ m}^3$. $T = 2.3 \text{ s}$, $A = 750 \text{ m}^2$

Reverberation time, T (*Sabine's formular for reverberation time*)

$$\alpha = \frac{0.158V}{AT} = \frac{0.158 \times 5500}{750 \times 2.3} = \mathbf{0.504}$$

4. The volume of a room is 600 m³. The wall area of the room is 220 m², the floor area is 120 m² and the ceiling area is 120 m². The average sound absorption coefficient, (i) for the walls is 0.03; (ii) for the floor is 0.06 and (iii) for the ceiling is 0.80. Calculate the average sound absorption coefficient α and the reverberation time.

Solution. Average absorption coefficient.

$$\alpha = \frac{\Sigma \alpha A}{\Sigma A} = \frac{\alpha_1 A_1 + \alpha_2 A_2 + \alpha_3 A_3}{A_1 + A_2 + A_3} = \frac{0.03 \times 220 + 0.06 \times 120 + 0.8 \times 120}{220 + 120 + 120} = 0.2387$$

Total sound absorption of the room.

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Reverberation time,
$$T = \frac{0.158V}{\alpha \Sigma A} = \frac{0.158 \times 600}{109.8} = 0.8634 \text{ s}$$

5. A hall of volume 5500 metre³ is found to have a reverberation time of 2.3 s. The sound absorbing surface of the hall has an area of 750 m². Calculate the average absorption coefficient.

Solution. Here, $V = 5500 \text{ m}^3$. $T = 2.3 \text{ s}$, $A = 750 \text{ m}^2$

Reverberation time, T (*Sabine's formular for reverberation time*)

$$\alpha = \frac{0.158V}{AT} = \frac{0.158 \times 5500}{750 \times 2.3} = 0.504$$

6. The volume of a room is 600 m³. The wall area of the room is 220 m², the floor area is 120 m² and the ceiling area is 120 m². The average sound absorption coefficient, (i) for the walls is 0.03; (ii) for the floor is 0.06 and (iii) for the ceiling is 0.80. Calculate the average sound absorption coefficient α and the reverberation time.

Solution. Average absorption coefficient.

$$\alpha = \frac{\sum \alpha A}{\sum A} = \frac{\alpha_1 A_1 + \alpha_2 A_2 + \alpha_3 A_3}{A_1 + A_2 + A_3} = \frac{0.03 + 220 + 0.06 \times 120 + 0.8 \times 120}{220 + 120 + 12} = 0.2387$$

Total sound absorption of the room.

$$\alpha \sum A = 0.2387 \times 460 = 109.8 \text{ metric sabine}$$

$$\text{Reverberation time, } T = \frac{0.158V}{\alpha \sum A} = \frac{0.158 \times 600}{109.8} = 0.8634 \text{ s}$$

7. Calculate the amplitude of the displacement wave, given that for a frequency of 400 hertz, the feeblest sound that could be heard corresponds to pressure amplitude of $8 \times 10^{-5} \text{ Nm}^{-2}$. The density of air is 1.29 kg m^{-3} and velocity of sound in air is 345 ms^{-1} .

$$\text{Sol. } a = \frac{P_{\max}}{2\pi\rho v n} = \frac{(8 \times 10^{-5})}{2 \times 1.29 \times 345 \times 400} = 7.15 \times 10^{-11} \text{ m}$$

Compare the acoustic intensities in air and in water for the same frequency and amplitude.

$$\text{Sol. } \frac{I_{\text{air}}}{I_{\text{water}}} = \frac{\rho v_{\text{air}}}{(\rho v)_{\text{water}}} = \frac{1.29 \times 350}{1000 \times 1470} = \frac{1}{3256}$$

5. Write a brief account of the acoustical analysis and its correction

Ans. Acoustic analysis is to check the prevailing condition of the hall for good acoustics. For this, it is necessary to determine the absorption and distribution of sound in the hall. If the absorption is inadequate then by adding or removing absorbing materials, the absorption in the hall is corrected. By adjusting the shape and interior surfaces of the hall, uniform sound distribution is ensured.

ULTRASONICS

The human ear is sensitive to sound waves in the frequency range from 20 to 20,000 Hz. This range is called audible range. Sound waves of frequency more than 20,000 Hz are called ultrasonics. These frequencies are beyond the audible limit.

These waves also travel with the speed of sound (330 ms^{-1}).

These waves exhibit the properties of audible sound waves and also show some new phenomena.

Their wavelengths are small.

PIEZOELECTRIC EFFECT

If one pair of opposite faces of a quartz crystal is subjected to pressure, the other pair of opposite faces develop equal and opposite electric charges on them (Figure 3). The sign of the charge is reversed when

the faces are subjected to tension instead of pressure. The electric charge developed is proportional to the amount of pressure or tension. This phenomenon is called piezoelectric effect.

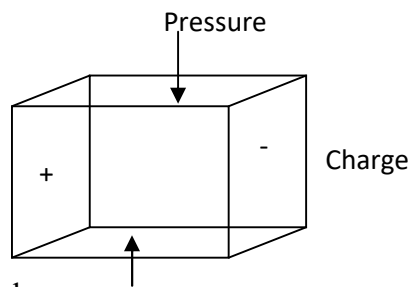


Figure 3: Quartz crystal

The effect is reversible i.e., if an electric field is applied across one pair of faces of the crystal, contraction or expansion occurs across the other pair.

When the two opposite faces of a quartz crystal, their faces being cut perpendicular to the optic axis, are subjected to alternating voltage, the other pair of opposite faces experiences stresses and strains. The quartz crystal will continuously contract and expand. Elastic vibrations are set up in the crystal.

When the frequency of the alternating voltage is equal to the natural frequency of vibration of the crystal or its simple higher multiples, the crystal is thrown into resonant vibrations. The amplitude is large. These vibrations are longitudinal in nature.

Consider a X-cut crystal plate of thickness t . The fundamental frequency of vibration is given by

$$n = \frac{t}{2r} \sqrt{\frac{E}{\rho}}$$

E is the Young's modulus and ρ is the density of the material of the crystal plate.

PRODUCTION OF ULTRASONIC WAVES – PIEZO-ELECTRIC CRYSTAL METHOD

Principle

This is based on the inverse piezoelectric effect. When a quartz crystal is subjected to an alternating potential difference along the electric axis, the crystal is set into elastic vibrations along its mechanical axis. If the frequency of the electric oscillations coincides with the natural frequency of the crystal, the vibrations will be of large amplitude. If the frequency of the electric field is in the ultrasonic frequency range, the crystal produces ultrasonic waves.

Construction

The circuit diagram is a base tuned oscillator circuit. A slice of quartz crystal is placed between the metal plates P and Q so as to form a parallel plate capacitor with the crystal as the dielectric. This is coupled to the electronic oscillator through primary coil L_1 of the transformer.

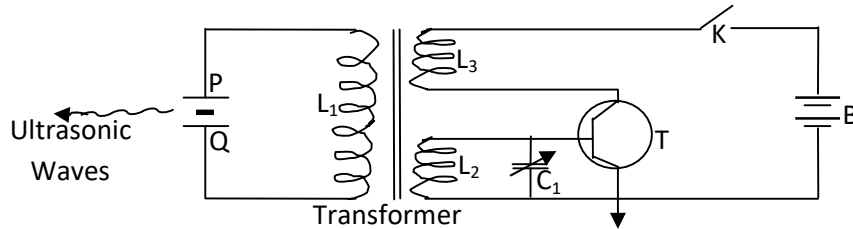


Figure 4: Circuit diagram of piezoelectric crystal ultrasonic generator

Coils L_2 and L_3 of oscillator circuit are taken from the secondary of the transformer. The collector coil L_3 is inductively coupled to base coil L_2 . The coil L_2 and variable capacitor C_1 form the tank circuit of the oscillator.

Working

When the battery is switched on, the oscillator produces high frequency oscillations. An oscillatory e.m.f. is induced in the coil L_1 due to transformer action. So the crystal is now under high frequency alternating voltage.

The capacitance of C_1 is varied so that the frequency of oscillations produced is in resonance with the natural frequency of the crystal. Now the crystal vibrates with large amplitude due to resonance. Thus high power ultrasonic waves are produced.

Advantages

1. Ultrasonic frequencies as high as 500 MHz can be generated.
2. The output power is very high. It is not affected by temperature and humidity.
3. It is more efficient than magnetostriction oscillator.
4. The breadth of the resonance curve is very small. So we can get a stable and constant frequency of ultrasonic waves.

Disadvantages

1. The cost of the quartz crystal is very high.
2. Cutting and shaping the crystal is very complex

PRODUCTION OF ULTRASONIC WAVES – MAGNETOSTRICTION METHOD

Principle

When a rod of ferromagnetic material like nickel is magnetized longitudinally, it undergoes a very small change in length. This is called magnetostriction effect.

A nickel rod placed in a rapidly varying magnetic field alternately expands and contracts with twice the frequency of the applied magnetic field. By adjusting the frequency of the alternating magnetic field to be equal to the natural frequency of longitudinal vibration of the rod, resonance is produced. Due to resonance, vibrations of large amplitude are produced in the rod. Ultrasonic waves are emitted from the ends of the rod if the frequency of the alternating magnetic field is more than 20KHz.

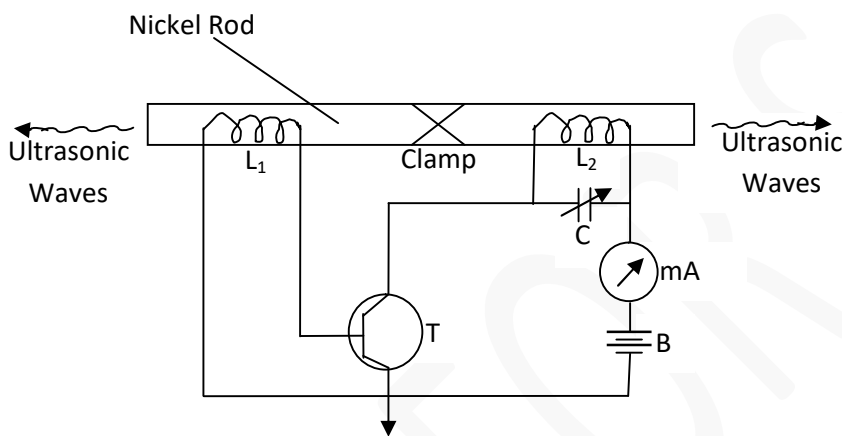


Figure 5: Circuit diagram of magnetostriction ultrasonic generator

The frequency of vibrations of the rod is

$$f = \frac{1}{2l} \sqrt{\frac{E}{\rho}}$$

Here, l = length of the rod,

E = Young's modulus of the material of the rod,

ρ = density of the material of the rod.

A short permanently magnetized nickel rod is clamped in the middle between two knife edges. A coil L_2 is wound on the right hand portion of the rod. C is a variable capacitor. L_2 and C form the resonant circuit of the collector tuned oscillator. Coil L_1 wound on the left hand portion of the rod is connected in the base circuit. The coil L_1 is used as a feedback loop.

Working

When the battery is switched on, the resonant circuit $L_2 C$ sets up an alternating current of frequency.

$$f = \frac{1}{2\pi\sqrt{L_2 C}}$$

This current flowing round the coil L_2 produces an alternating magnetic field of frequency f along the length of the nickel rod. The rod starts vibrating due to magnetostrictive effect. The vibrations of the rod create ultrasonic waves.

The longitudinal expansion and contraction of the rod produces an e.m.f in the coil L_1 . This e.m.f is applied to the base of the transistor. Hence the amplitude of high frequency oscillations in coil L_2 is increased due to positive feedback.

The developed alternating current frequency can be tuned with the natural frequency of the rod by adjusting the capacitor. The resonance condition is indicated by the rise in the collector current shown in the milliammeter.

Advantages

1. Magnetostriction oscillators are mechanically rugged.
2. The construction cost is low.
3. They are capable of producing large acoustical power with fairly good efficiency (e.g. 60%).

Limitations

1. It can produce frequencies up to 3 MHz only.
2. The frequency of oscillations depends greatly on temperature.
3. Breadth of the resonance curve is large. It is due to variations of elastic constants of ferromagnetic material with the degree of magnetisation. So we cannot get a constant single frequency.

DETECTION OF ULTRASONIC WAVES

1. Quartz crystal method

This method is based on the principle of piezoelectric effect. When one pair of opposite faces of a quartz crystal is exposed to the ultrasonic waves, the other pair of opposite faces develop opposite charges. These charges are amplified and detected using an electronic circuit.

2. Thermal detectors

A platinum wire is placed in the region to be tested for ultrasonic waves. At nodes, due to alternate compressions and rarefactions, alternate heating and cooling is produced. Change in temperature at

the node brings about changes in the electrical resistance of the platinum wire. This is detected by means of a Whetstone's bridge. No change in temperature occurs at the antinodes.

3. **Kundt's tube method**

If the wavelength of the ultrasonic waves is greater than a few millimeters, a Kundt's tube can be used to form a stationary wave pattern with well defined nodes and antinodes. In the air or gas medium, lycopodium powder collects in the form of heaps at the nodes. In liquid medium, powdered coke is used to detect the position of nodes.

PROPERTIES OF ULTRASONIC WAVES

1. They have high energy content.
2. Just like ordinary sound waves, ultrasonic waves get reflected, refracted and absorbed.
3. They show negligible diffraction due to their small wavelength. Hence they can be transmitted over long distances without any appreciable loss of energy.
4. They produce intense hearing effect when passed through a substance.
5. If an arrangement is made to form stationary waves of ultrasounds in a liquid, it serves as a diffraction grating. It is called an acoustic grating.

APPLICATIONS OF ULTRASONIC WAVES

(i) Science

1. **Investigation of Structure of Matter**
2. **Study of Molecular Energies**
3. **Elastic Symmetries of Crystals**
2. **Ultrasonic Soldering**
3. **Ultrasonic Welding**
4. **Ultrasonic Drilling and Cutting**
5. **Ultrasonic in Metallurgy**
6. **Formulation of Alloys**
7. **Acoustic Holograms**
8. **Sound Navigation and Ranging (SONAR)**
9. **Chemical Applications**

1. Ultrasonic waves are used to form stable emulsions of even immiscible liquids like Water and oil or water and mercury. This finds an application in the preparation of photographic films, face creams etc.
2. They are used to liquefy gels like aluminium hydroxide in the same way as they are liquefied by shaking.
3. They are used to coagulate fine solid or liquid particles in a gas; for example, dust, smoke, mist etc. Ultrasonic thus find use in collecting factory dust and purifying the air.
4. Ultrasonics act like a catalytic agent and accelerate chemical reactions. Ultrasonic waves accelerate crystallization.

(iii) **Medical Applications**

1. **Disease treatment**
2. **Surgical use**
3. **Diagnostic use**
4. **Extraction of broken teeth**
5. **Sterilization**
6. **Blood flow meters**

HEARING

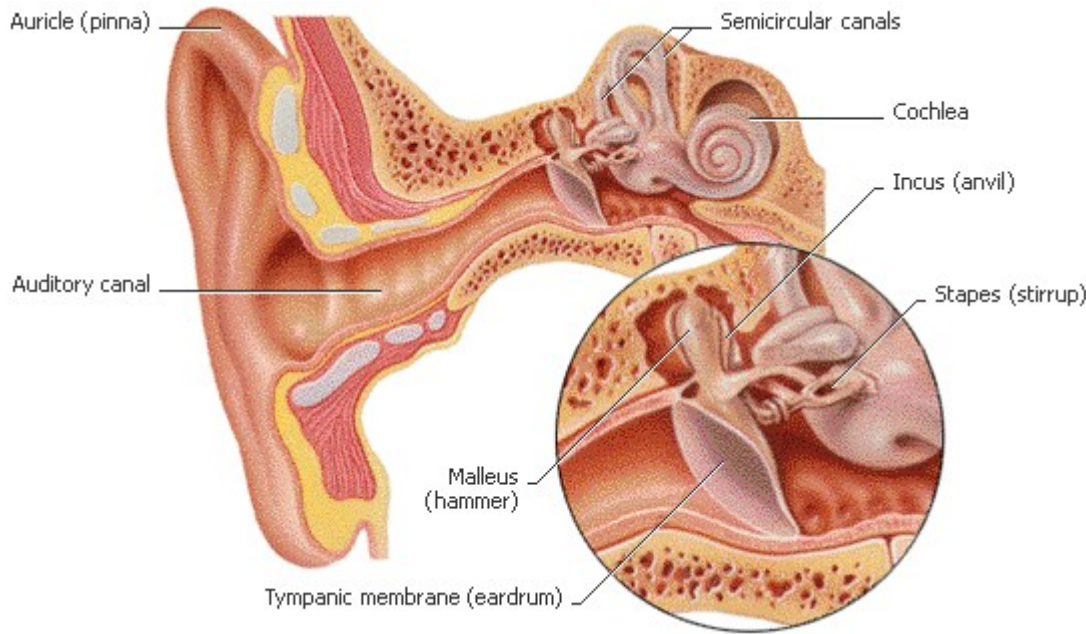


Figure 6: Structure of the Ear

The human ear consists of three sections: the outer ear, the middle ear, and the inner ear. The outer ear includes the auricle (pinna), the visible part of the ear that is attached to the side of the head, and the waxy, dirt-trapping auditory canal. The tympanic membrane (eardrum) separates the external ear from the middle ear, an air-filled cavity. Bridging this cavity are three small bones—the malleus (hammer), the incus (anvil), and the stapes (stirrup). The cochlea and semicircular canals make up the inner ear.

Figure 6 gives a diagram of the human ear. When a sound wave arrives at human ear, the outer ear collects the wave and transmits it to the middle ear through the eardrum. The bones in the middle ear control the amplitudes of the vibrations transmitted to the inner ear. Nerves in the cochlea of the inner ear respond to the sound wave received and transmit the moderated vibrations to the brain. The brain process and interpret the sound waves.

HOW MOUTHPIECE AND EARPIECE WORK

LOUDSPEAKER

The Loudspeaker consists of diaphragm and an electromagnet. When the small current from the microphone flow through the coil of the electromagnet, the magnetic flux changes and the diaphragm is set into vibration. Thereby converting electric energy into sound energy.

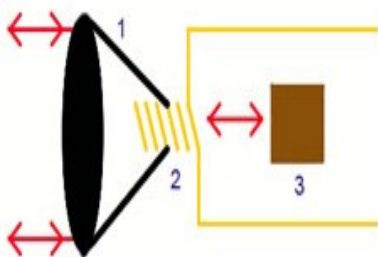


Figure 7: Schematic diagram of speakers

Inside a speaker:

1. Cone
2. Electromagnet (coil)
3. Permanent magnet

The frequency of the vibrations governs the pitch of the sound produced, and their amplitude affects the volume.

MICROPHONE

A microphone uses the same mechanism as a speaker in reverse to convert sound into an electrical signal. When a person speaks before the microphone, the diaphragm vibrates, and the pressure on the Carbon granules changes continuously. The resistance offered by the carbon granules in the circuit also changes. When they are loosely packed, the resistance increases and the current in the circuit decreases thereby converting the sound energy to small electric current.

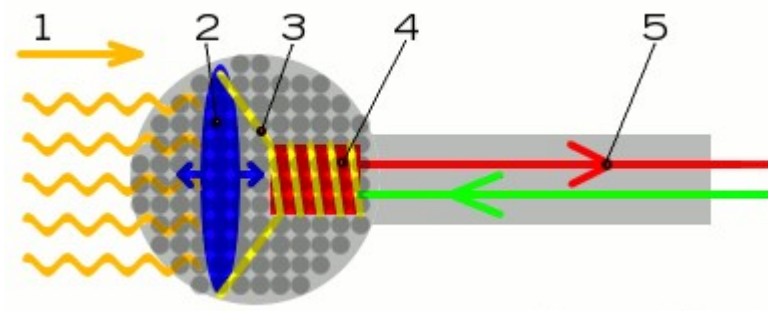


Figure 8: How microphone works